

Chapter 1

Operating System

→ Operating systems are those system programs that controls the available resources of a computer system and also provides the interface between the hardware and other application programs.

→ An operating system is a program that acts as an interface between the user and the computer hardware and controls the execution of all kinds of programs.

→ Some examples of OS are Linux, Windows, OSX, Solaris, OS/400, etc.

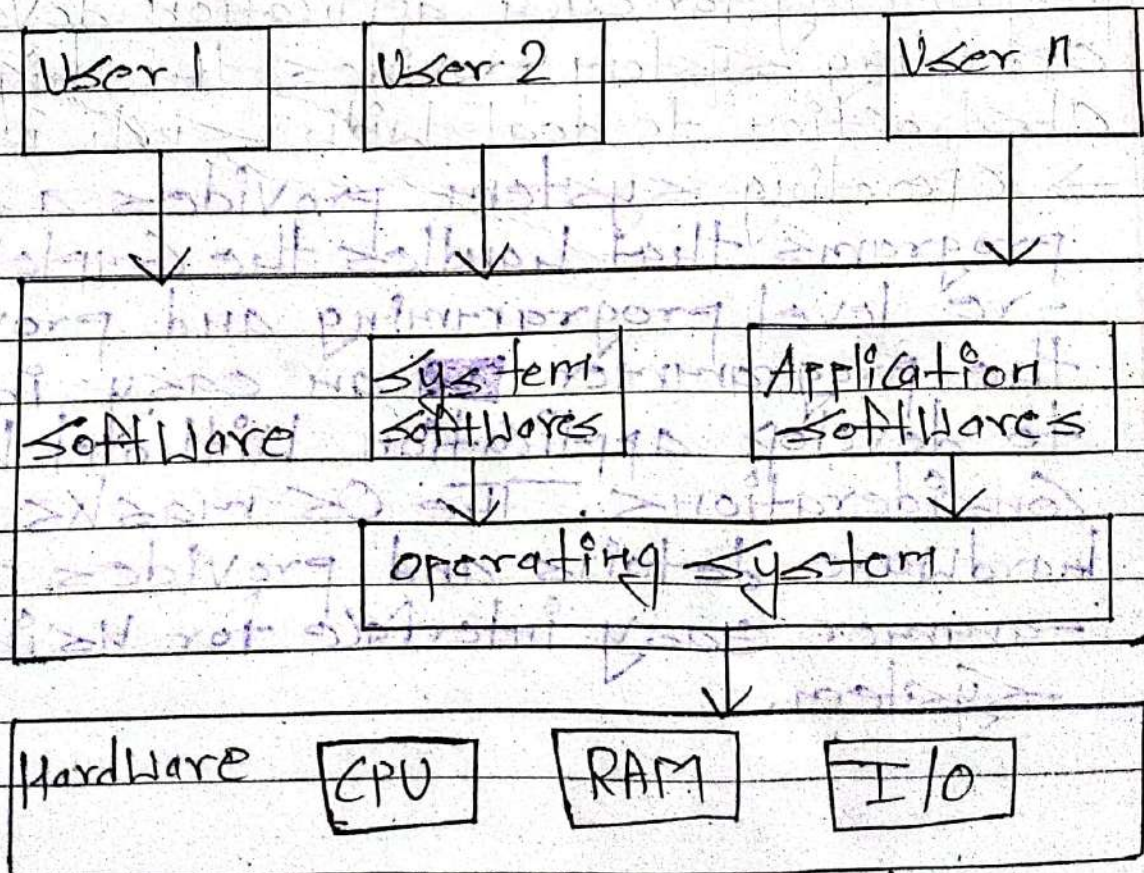


Fig. Operating System

- Operating system performs two basic functions:
- i) Resource management
 - ii) Virtual machine management
- Services provided by OS

Operating System (OS) as extended or virtual machine or mediator

→ The computer system is composed of hardware and application software interacting with each other to perform any computation. For any application to perform its task, it must deal with hardware for memory management, process control, storage and so on. But for a programmer, it would be tedious to consider the hardware level machine language programming for each application development. Operating system provides the high level abstraction to deal with such issues.

→ Operating system provides a set of programs that handles the complex hardware level programming and provides the programmer an easy interface to develop application without hardware considerations. The OS masks the hardware details and provides the programmer easy interface for using the system.

Operating System Provides Abstraction. Justify

→ For eg: Consider the case of disks. A disk contains collection of files. Each files can be opened, read, written and closed. The hardware details like it regarding should use, modified FM or the current state of motor, etc are masked by OS and also handled effectively.

OS as resource manager

→ A Computer system is a system with various resources like memory, processor, disk, printer, etc. It is necessary to keep track of the resource allocation and usage in order to resolve conflict between two or more processes for allocation of same resources. Also, in a networked system, the sharing of hardware and information is very common. So, the need of managing memory and resources among the processes or the users is necessary.

→ The operating system provides an orderly and controlled allocation of resources among the programs or the users. It keeps track of current usage status of resources, who is using resource, to grant resource request if available and so on.

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→ For eg: If 3 programs are requesting for a printer service, the OS schedule the tasks based on priority. It can be managed by buffering all the output destined for the printer on disk.

objective/goals of OS / Purpose of OS

→ An operating system has a special program in which controls the execution of applications or programs. OS acts as an intermediary among applications and the hardware components. OS can be thought of as having three objectives. These are:

i) Convenience

It makes a computer more suitable to use.

ii) Efficiency

It provides the computer system resources with efficiency and in easy to use format.

iii) Ability to develop

It should be built in such a way that it permits the efficient development, testing and installation of new system functions without interfering with services.

Objective: i) To hide details of hardware by creating abstraction

ii) To allocate resources to processes (manage resources)

iii) provide pleasant and effective user interface

Services provided by Operating System

→ An Operating System provides services to both the users and to the programs.

i) It provides programs an environment to execute.

ii) It provides users the services to execute the programs in a convenient manner.

few common

→ The services provided by an Operating System are:

i) Program Execution

The operating system must have the capability to load a program into ~~memory~~ memory and execute that program. Furthermore, the program must be able to end its execution, either normally or abnormally/forcefully.

ii) File System Manipulation

Programs need data to be read and then write them as files and directories. File handling portion of operating system also allows users to create and delete files by specific name along with extension, search for a given file and/or

list file information. Some programs comprise of permissions management for allowing or denying access to files or directories based on file ownership.

iii) I/O operations

A program which is currently executing may require I/O, which may involve file or other I/O device. For efficiency and protection, users cannot directly govern the I/O devices. So, the OS provide a means to do I/O ~~input~~ operation which ~~means~~ means read or write operation with any file.

iv) Communication ~~system~~

process needs to swap over information with other process. Processes executing on same computer system or on different computer systems can communicate using operating system support. Communication between two processes can be done using shared memory or via message passing.

v) Resource Allocation

When multiple jobs running concurrently, resources must need to be allocated to each of them. Resources can be CPU

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cycle, main memory storage, file storage and I/O devices. CPU scheduling routines are used here to establish how best the CPU can be used.

Vi) Error Detection

Errors may occur within CPU, memory hardware, I/O devices and in the user program. For each type of error, the OS takes adequate action for ensuring correct and consistent computing.

Vii) Accounting

This service of OS keeps track of which users are using how much and what kinds of computer resources have been used for accounting or simply to accumulate usage statistics.

Viii) Protection and Security

Protection includes in ensuring all access to system resources in a controlled manner. For making a system secure, the user needs to authenticate him or her to the system before using.

Functions of Operating System:

Following are some of important functions of an operating system:

i) Processor management

It involves putting the tasks into order and pairing them into manageable size before they go to the CPU.

ii) Memory management

It ~~coordinates~~ coordinates data to and from RAM and determines the necessity for virtual memory.

iii) Device management

It provides interface between connected devices.

iv) Storage management

It directs permanent data storage.

v) Application

It allows standard communication between software and our computer.

vi) User Interface

It allow you to communicate with your computer.

Vii) It provides file management which refers to the way that the operating system manipulates, stores, retrieves and saves data.

Viii) Error Handling is done by operating system. It takes preventive measures whenever required to avoid errors.

ix) Security - By means of password and similar other techniques, it prevents unauthorized access to program and data.

X) Control over system performance
Recording delays between request for a service and response from the system.

xi) Coordination between other softwares and users:

Coordination and assignment of compilers, interpreters, assemblers and other software to the various users of computer systems.

Kernel mode and user mode

In any modern operating system, the CPU is actually spending time in two very distinct modes to make sure it works correctly:

* Kernel mode

- In kernel mode, the executing code has complete and unrestricted access to the underlying hardware. It can execute any CPU instruction and reference any memory address. Kernel mode is generally reserved for the lowest-level, most trusted functions of the operating system.
- Crashes in kernel mode are catastrophic, they will halt the entire PC.

* User Mode

- In user mode, the executing code has no ability to directly access hardware or reference memory.
- Code running in user mode must delegate to system APIs to access hardware or memory. Due to the protection afforded by this sort of isolation, crashes in user mode are always recoverable.
- Most of the code running on your computer will execute in user mode.

Similarities and Difference between User mode and Kernel mode

Similarities:

→ The computer can switch between both modes.

Differences:

Basis	Kernel mode	User mode
Permission	Unrestricted and full permissions to access the system's hardware.	Restricted and limited permissions to access the system's hardware.
Memory reference	It can reference to both the memory spaces.	It can only reference to memory space that is dedicated to user mode.
Access	only core functionality can be allowed to operate in this mode.	User applications can access this mode for a particular system and is allowed to operate in this particular mode.
System crash	Fatal and increases the complexity.	Recoverable and can simply restart the session.
Also known	privileged mode or supervisor mode or master mode	Restricted mode or slave mode

Necessity of Dual Mode (User Mode and Kernel Mode) in OS

The lack of dual mode i.e. user mode and kernel mode in an operating system can cause serious problems. Some of them are:

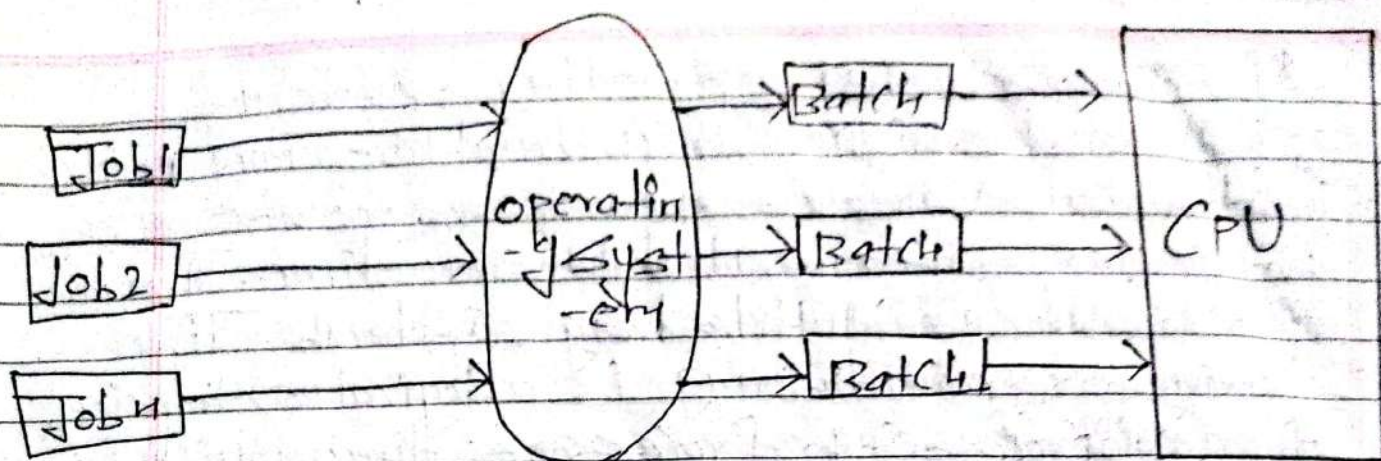
- 1) A running user program can accidentally wipe out the operating system by overwriting it with user data.
- 2) Multiple processes can write in the same system at the same time, with disastrous result.

To avoid such problems dual mode are necessary in OS.

Types of OS

Some of the important types of operating system which are most commonly used are:

- 1) Batch operating system
→ This type of operating system do not interact with computer directly. There is an operator which takes similar jobs having same requirement and group them into batches. It is the responsibility of operator to sort the jobs with similar needs.



Advantages:

- It is very difficult to guess or know the time required by any job to complete. Processor of the batch systems know how long the job would be when it is in queue.
- Multiple users can share the batch systems.
- The idle time batch system is very less.
- It is easy to manage large work repeated -ly in batch systems.

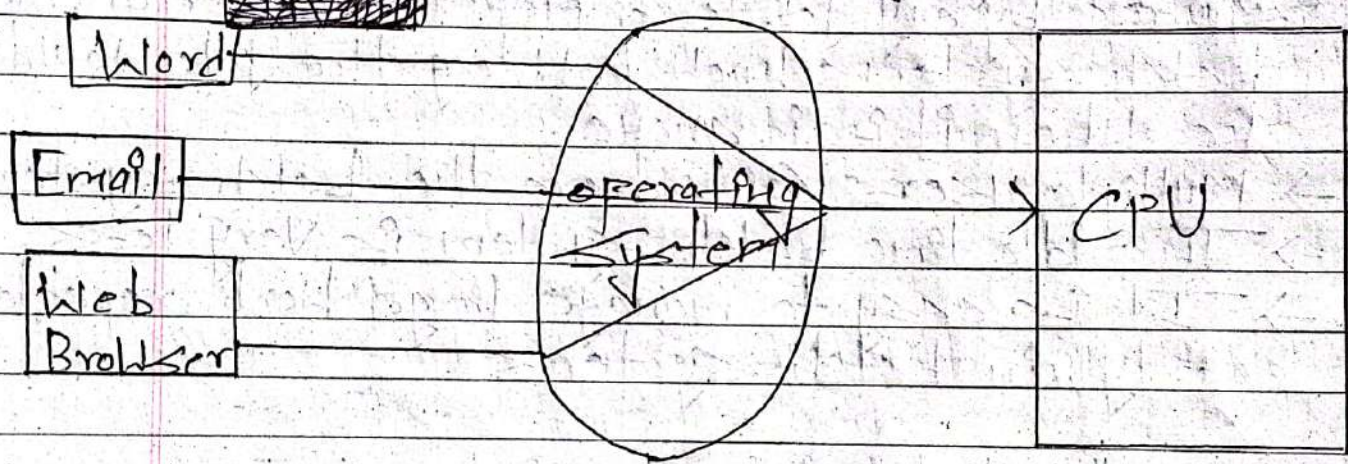
Disadvantages:

- The computer operators should be well known with batch systems.
- Batch systems are hard to debug.
- It is sometime costly.
- The other jobs will have to wait for an unknown time if any job fails.

Example: payroll system, Bank statements etc.

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ii) Time sharing operating systems
It is a technique which enables many people located at various terminals, to use a particular system at the same time. They are also known as multitasking systems. Time sharing or multitasking is a logical extension of multiprocessing. Processor's time which is shared among multiple user simultaneously is termed as time sharing.



Advantages:

- Each task gets an equal opportunity
- Less chances of duplication of software
- CPU idle time can be reduced

Disadvantages:

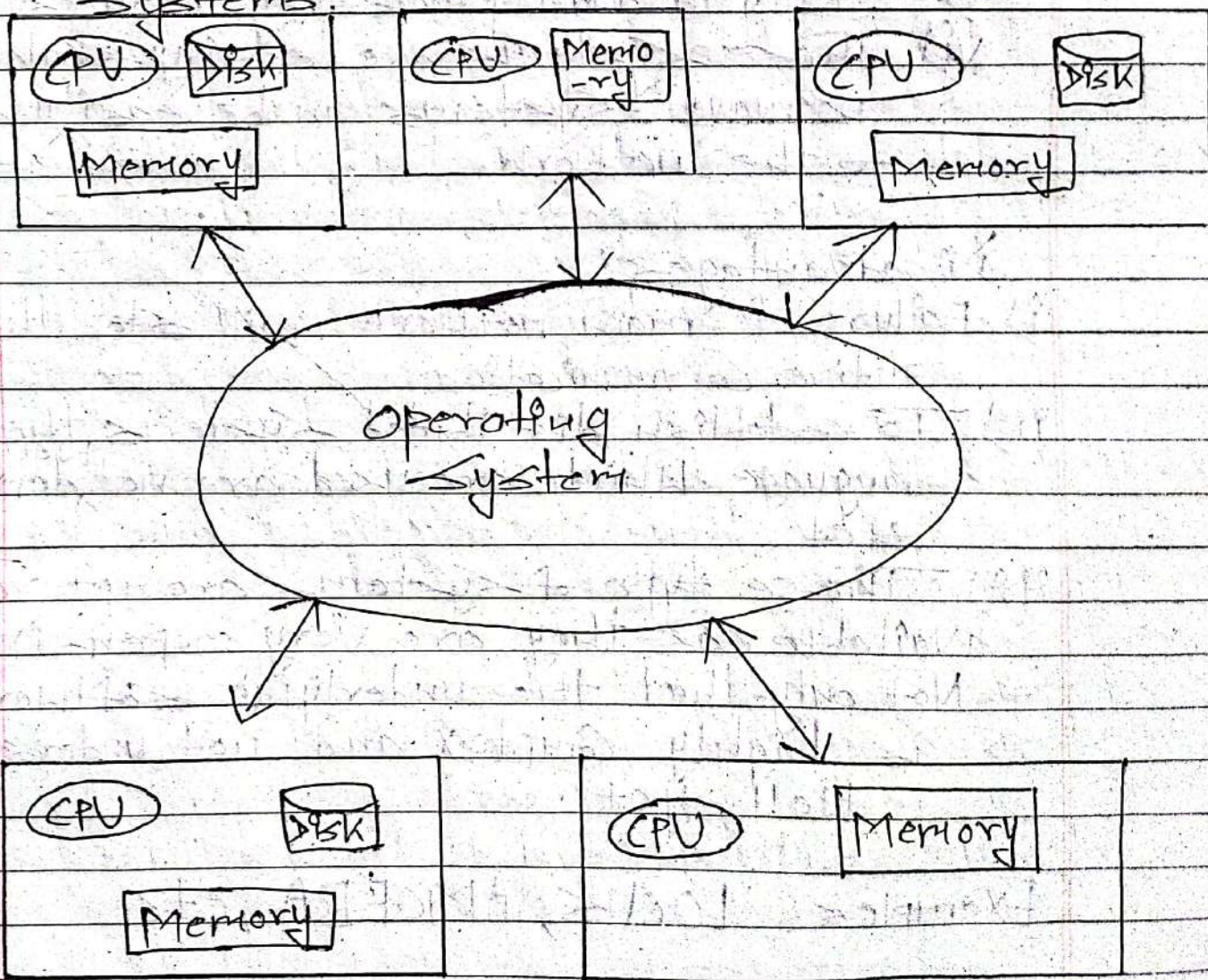
- Reliability problem
- Question of security and integrity of user program and data
- problem of data communication

Example: Multics, Unix etc.

iii) Distributed operating system

→ A distributed OS is an operating system that runs on several machines. Its purpose is to provide a useful set of services, generally to mask the collection of machines behave more like a single machine.

→ Independent systems possess their own memory unit and CPU. These are referred as loosely coupled systems or distributed systems.



Advantages:

- i) Failure of one will not affect the other network communication, as all systems are independent from each other.
- ii) Electronic mail increases the data exchange speed.
- iii) Since resources are being shared, computational is highly fast and durable.
- iv) Load on host computer reduces.
- v) Delay in data processing reduces.
- vi) These systems are easily scalable as many systems can be easily added to the network.

Disadvantages:

- i) Failure of main network will stop the entire communication.
- ii) To establish distributed systems the language which are used are not ^{well} defined yet.
- iii) These type of systems are not readily available as they are very expensive. Not only that the underlying software is highly complex and not understood well yet.

Examples: LOCUS, AMOEBA, etc.

iv) Network Operating System

This system runs on a server and provides the server the capability to manage data, users, groups, security, applications, and other networking functions. The primary purpose of the network operating system is to allow shared files and printer access among multiple computers in a network.

It is further divided in two types:

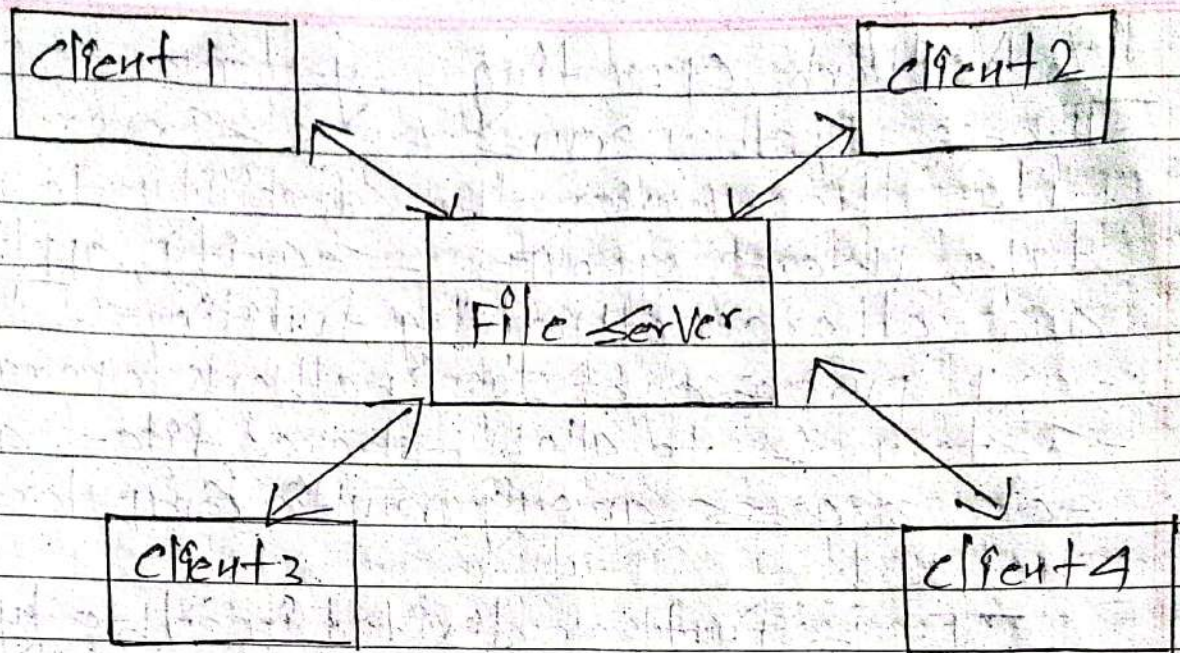
a) Peer-to-Peer

It allows users to share resources and files located on their computers and to access shared resources found on other computers.

b) Client/Server

It allows the network to centralize functions and applications in one or more dedicated file servers.

Examples: Novell NetWare, Windows 2000 Server, Windows Server 2008, UNIX, Linux, BSD, etc.



Advantages:

- Highly stable Centralized Servers
- Security concerns are handled through servers.
- New technologies and hardware up-gradation are easily integrated to the system.
- Server access are possible remotely from different locations and types of systems.

Disadvantages:

- Servers are costly
- User has to depend on central location for most operations
- Maintenance and updates are required regularly.

V) Parallel operating system

- parallel operating systems are used to interface multiple networked computers to complete task in parallel.
- The architecture of the software is often a UNIX-based platform, which allows it to coordinate distributed loads between multiple computers in a network.
- parallel operating systems are able to use software to manage all of the different resources of the computers running in parallel, such as memory, caches, storage space, and processing power.
- parallel operating systems also allow a user to directly interface with all of the computers in network.
- A parallel operating system works by dividing sets of calculations into smaller parts and distributing them between the machines on a network.

Vi) Real Time operating system

- A real time system is defined as data processing system in which the time interval required to process and respond to inputs is so small that it controls the environment.
- It is a multitasking operating system that aims at executing real-time application
- A real time operating system must have

Well-defined, fixed time constraints, otherwise the system will fail.

→ For example, Scientific experiments, medical imaging system, industrial control systems, weapon systems, robot, air traffic control system, etc.

→ It is of two types:

i) Hard real-time systems

Hard real-time systems guarantee that critical tasks complete on time. In this system secondary storage is limited or missing and the data is stored in ROM.

ii) Soft real-time systems

→ They are less restrictive.

→ They have limited utility than hard-real time system

Advantages:

→ Maximum consumption

→ Task shifting

→ Error free

→ 24-7 systems

Disadvantages:

→ Limited tasks

→ Use heavy system resource

→ Complex Algorithms

→ Expensive

Structure of operating system:

An operating system might have many structure. Some of the main structures used in operating system are:

i) Monolithic architecture

- It is the oldest architecture used for developing operating system.
- Here, the kernel is a single large program.
- Functionality of the OS is invoked with simple function calls with the kernel.
- Device drivers are loaded into the running kernel and become part of the kernel.
- System calls from the user programs are kept in trap table, which are executed in kernel mode switching from user mode.

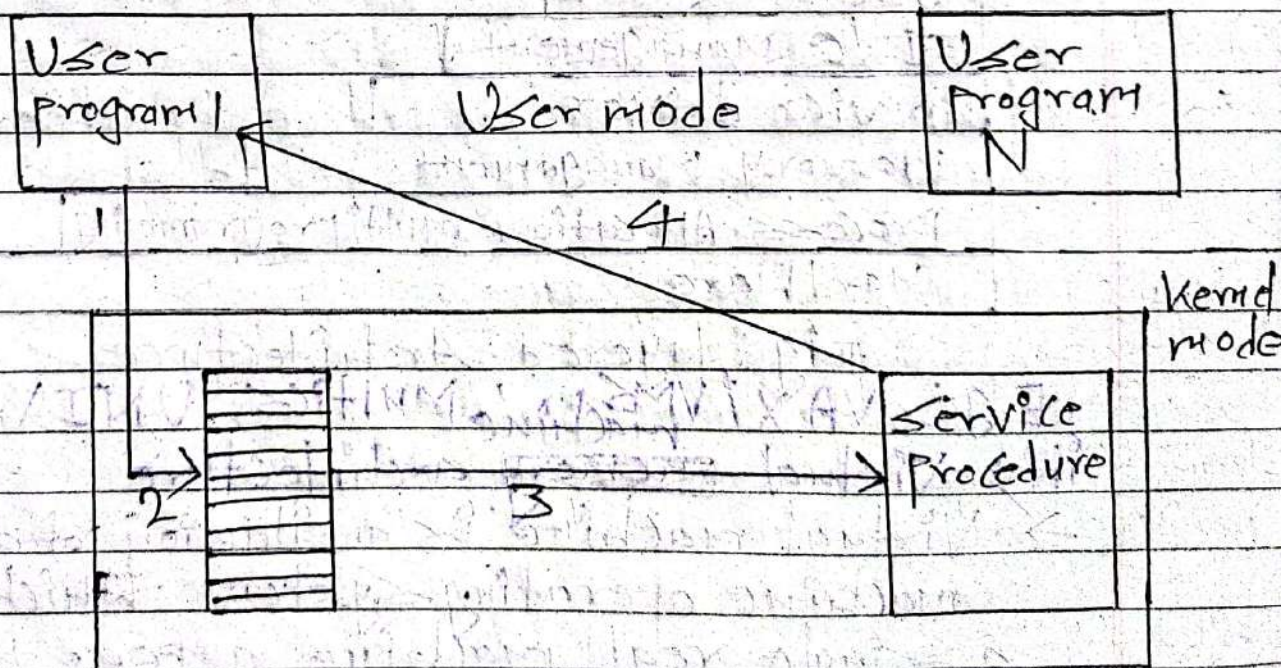


Fig: monolithic structure of OS
→ Eg: CPM and MS-DOS

ii) Layered structure

- In this approach the operating system is broken up into number of layers.
- Hierarchy of layers, each constructed upon another below it.
- Layer 0 (bottom layer) is the hardware layer to layer N (highest layer) is the user interface layer.
- provide modularity
- With modularity, layers are selected such that each uses functions (operations) and services of only lower-level layer.
- But difficult in differentiation and less efficient.
- E.g: OS/2 operating system, earlier version of Windows NT

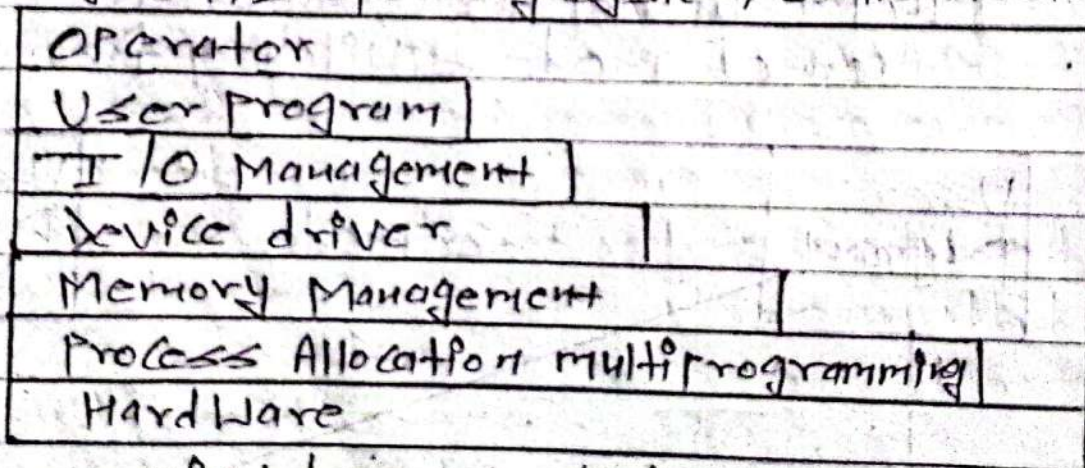


Fig: Layered Architecture

E.g: VAX/VMS, Multics, UNIX

iii) Virtual ~~memory~~ architecture

- Virtual machine is an illusion of a real machine operating system, which make a single real machine appear to be several real machine.

- In this system, each user can choose a different OS.
- Example: IBM 370 Computer, Java

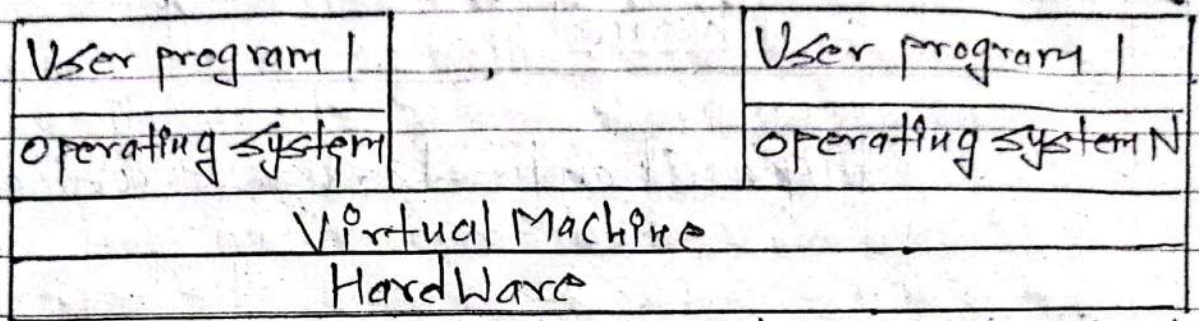


Fig: Virtual machine architecture

Advantages:

- Complete protection of system resource
- No direct sharing of resources

Disadvantages:

- Difficult to implement

iv) Client - Server ^{or Microkernel} architecture

- Two classes of processes - server & client
- Communication between client and server is via message passing
- Client and server can run on different computers connected by LAN/WAN
- Servers run as user mode. Hence no system down if the servers crashed.
- Well adapted in distributed system.

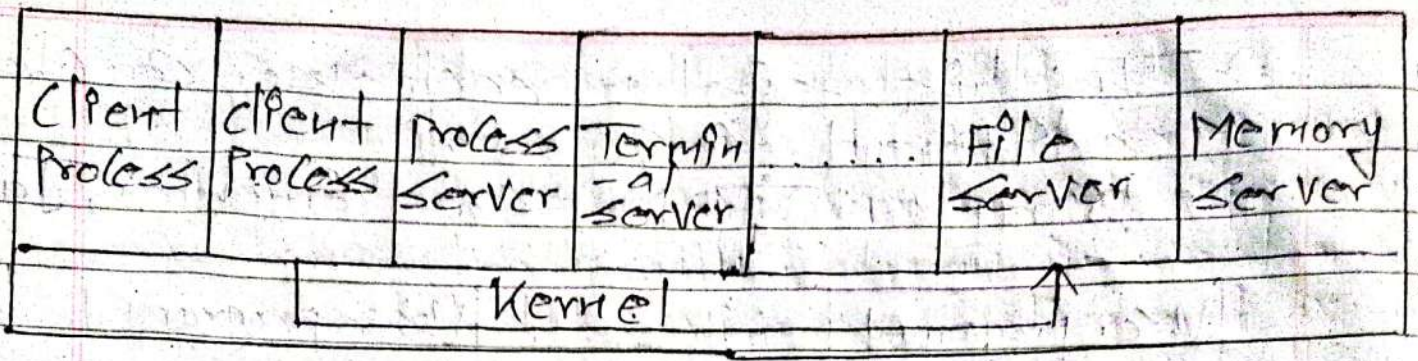


Fig: Client-server architecture

Difference Between Multitasking, Multiprogramming, Multiprocessing

Multiprogramming:

When multiple programs execute at a time on a single device, it is multiprogramming.

Multitasking:

When a single resource is used to process multiple tasks then it is multitasking.

Multiprocessing:

When more than one processing unit is used by a single computer then it is called multiprocessing.

Multiprogramming

1. In multiprogramming, multiple programs execute at a same time on a single device.

2. The process resides in the main memory.

3. It uses batch OS. The CPU is utilized completely while execution.

4. The processing is slower as a single job resides in the main memory while execution.

Multitasking

In multitasking, a single resource is used to process multiple tasks.

The process resides in the same CPU.

It is time sharing as the task assigned switch regularly.

Multitasking follows the concept of context switching.

Multiprocessing

In multiprocessing, multiple processing units are used by a single device.

The process switches from one to another CPU as multiple processing units are used.

It carries multiple processors to execute the task.

A large amount of work can be done in a short period of time.

System Calls

→ A System Call is how a program requests to a service from an operating system's kernel. This may include hardware-related services, creating and executing new processes and communicating with integral kernel services. System calls provide an essential interface between a process and the operating system. System calls are the only entry points into the kernel system. All programs needing resources must use system calls.

→ The services provided by system calls are: (Services provided)

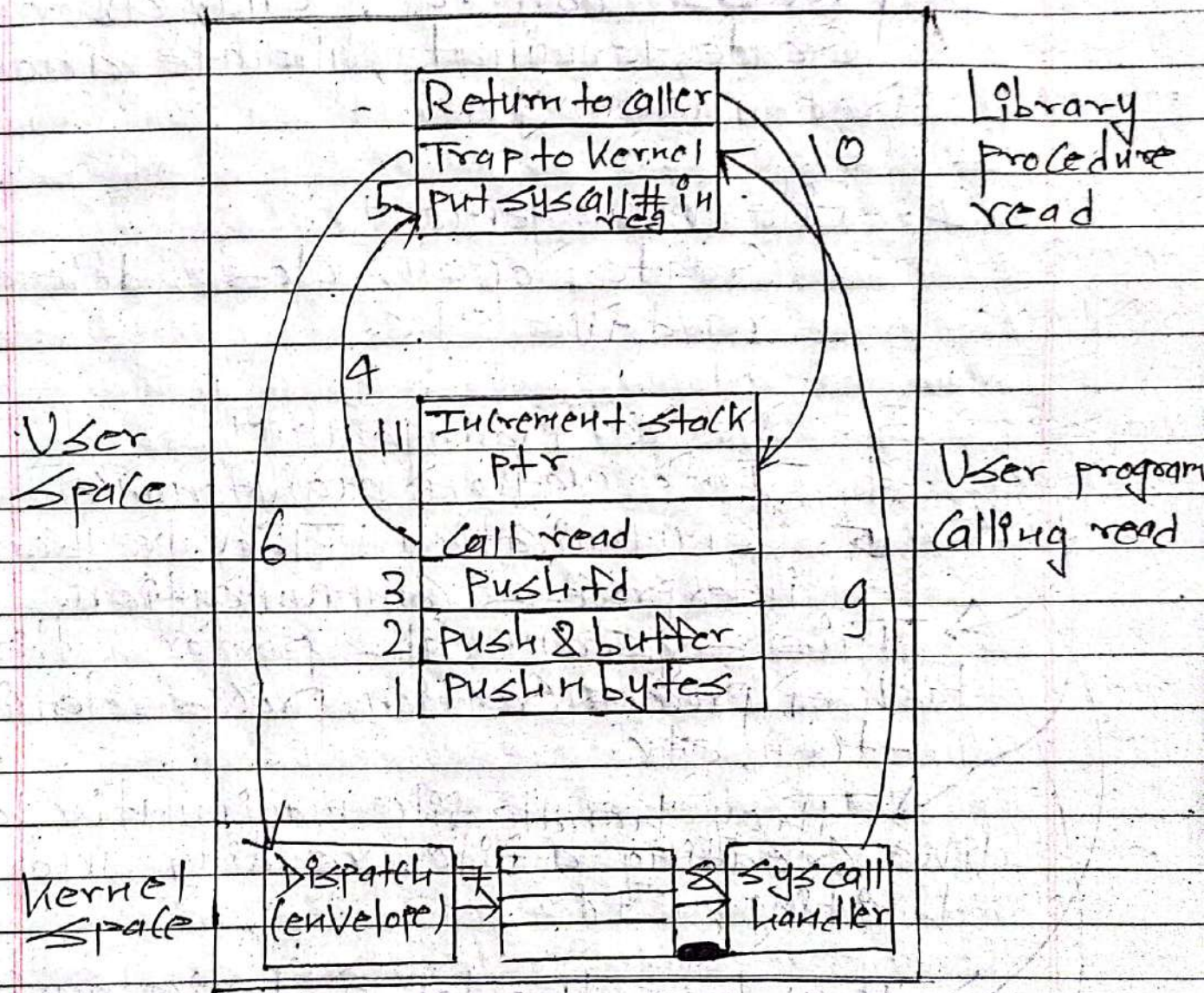
1. Process creation and management
2. Main memory management
3. File Access, Directory and File system management
4. Device Handling (I/O)
5. Protection
6. Networking, etc.

The working mechanism of system call is below
Example of read() system call

read(fd, buffer, nbytes)

The system call returns the number

of bytes actually read in `count`. This value is normally the same as `n` bytes, but may be smaller if, for example, end-of-file is encountered while reading. It is explained in below diagram:



Types of System Calls

There are 5 different Categories of System Calls:

1. Process Control: end, abort, Create, terminate, allocate and free memory.

2. File management: Create, open, close, delete, read file etc.

3. Device management

4. Information maintenance

5. Communication: They are used for interprocess communication.

→ This system calls perform the task of process creation, process termination, etc.

→ They handle file manipulation jobs like creating a file, reading and writing, etc.

→ They do the job of device manipulation like reading from device buffers, writing into device buffers, etc.

→ It handles information and its transfer between the OS and the user program.

History of operating system

i) The 1940's - First Generation

The earliest electronic digital computers had no operating systems. Machines of the time were so primitive that programs were often entered one bit at time on rows of mechanical switches (plug boards). Programming languages were unknown (not even assembly language). Operating systems were unheard of.

ii) The 1950's - Second Generation

By the early 1950's, the routine had improved somewhat with the introduction of punch cards.

The General Motors Research Laboratories implemented the first operating systems in early 1950's for their IBM 701.

The system of the 50's generally ran one job at a time.

iii) The 1960's - Third generation

The system of the 1960's were also batch processing systems, but they were able to take better advantage of the computer's resources by running several jobs at once.

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iv) Fourth Generation

With the development of LSI (Large Scale Integration) Circuits, Chips, operating systems entered in the personal computer and the Workstation age. Microprocessor technology evolved to the point that it becomes possible to build desktop computers as powerful as the mainframe of the 1970s.

Views of operating system (2 types) ^{mainly} views on

The operating systems are given as :

a) User ~~View~~ View

The user view depends on the system interface that is used by the users.

→ If the user is using a personal computer the operating system is largely designed to make interaction easy.

→ If the user is using a system connected to a mainframe computer or a minicomputer, the operating system is largely connected with resource ~~utilization~~ utilization.

→ If the user is sitting on a workstation connected to other workstations through networks, then OS needs to focus on both individual usage of resources and sharing through the network.

→ If the user is using a handheld computer such as a mobile then the OS handles the usability of the device including some remote applications.

ii) System View

According to the computer system, the OS is the bridge between application and hardware.

→ The system view the OS as a resource allocator.

→ The OS can also work as a control program.

→ OS can also be viewed as a way to make hardware easier.

→ OS were developed to easily communicate with hardware.

Operating System Components

The components of operating system are as follow:

i) Process management (scheduling)

→ A process is the unit of work in a system.

→ operating system manages the execution of those process.

→ OS is responsible for the following activities in connection with process management

x) The creation and deletion of both user and system processes.

- Suspension and
- * The ~~Supervision~~ resumption of processes.
 - * The provision of mechanism for process synchronization.
 - * The provision of mechanism for deadlock handling.
 - * Provision of mechanism for process communication.
 - ii) I/O management

one of the important job of OS is to manage various I/O devices. I/O system requires to take an application I/O request and send it to physical device, then take whatever response comes back from the device and send it to application.

iii) Main memory management
Main memory is a large array of words or bytes. Each word or byte has its own address. Main memory is a repository of quickly accessible data shared by the CPU and I/O devices.

The major activities of an operating system in regard to memory management are:

- i) Keep track of which part of memory are currently being used and by whom
- ii) Decide which processes are loaded into memory when memory space

becomes available.

iii) Allocate and deallocate memory space as needed.

iv) File management

→ A file is a collection of related information defined by its creator. Computer can store files on the disk (Secondary storage), which provide long-term storage.

→ The activities of OS in regard to file management are:

- * The Creation and deletion of files

- * The " " " " of directories

- * The support of primitives for manipulating files and directions.

- * The mapping of files onto Secondary storage.

- * The backup of files on stable storage media.

v) Secondary storage management

The operating system is responsible for the following activities in connection with disk management.

- Free space management

- Storage allocation

- Disk scheduling

Vi) Networking

A distributed system is a collection of processors that do not share memory, peripheral devices, or a clock. The processors communicate with one another through communication lines called network.

Vii) Protection System

Protection refers to mechanism for controlling the access of programs, processes, or user to the resources defined by a computer system.

Viii) Command Interpreter System

A command ~~interpreter~~ interpreter is a interface of the operating system with the user. The user gives commands which are executed by operating system.

operating system concepts

Process

- A ~~program~~ process is an instance of a program in execution
- A program becomes process when executable file is loaded in the main memory
- A process is defined as an entity which represents the basic unit of

Work to be implemented in the system.

ii) Files

- A file is a named collection of related information that is recorded on secondary storage such as magnetic disks, magnetic tapes and optical disks.
- In general, a file is a sequence of bits, bytes, lines or records whose meaning is defined by the file creator and the user.

iii) System Calls

See from previous description

iv) The Shell

- The shell is the outermost layer of the operating system. Shell incorporate a programming language to control the processes and files, as well as to start and control other programs.
- Shell provides a way for you to communicate with operating system.

Shell and Kernel

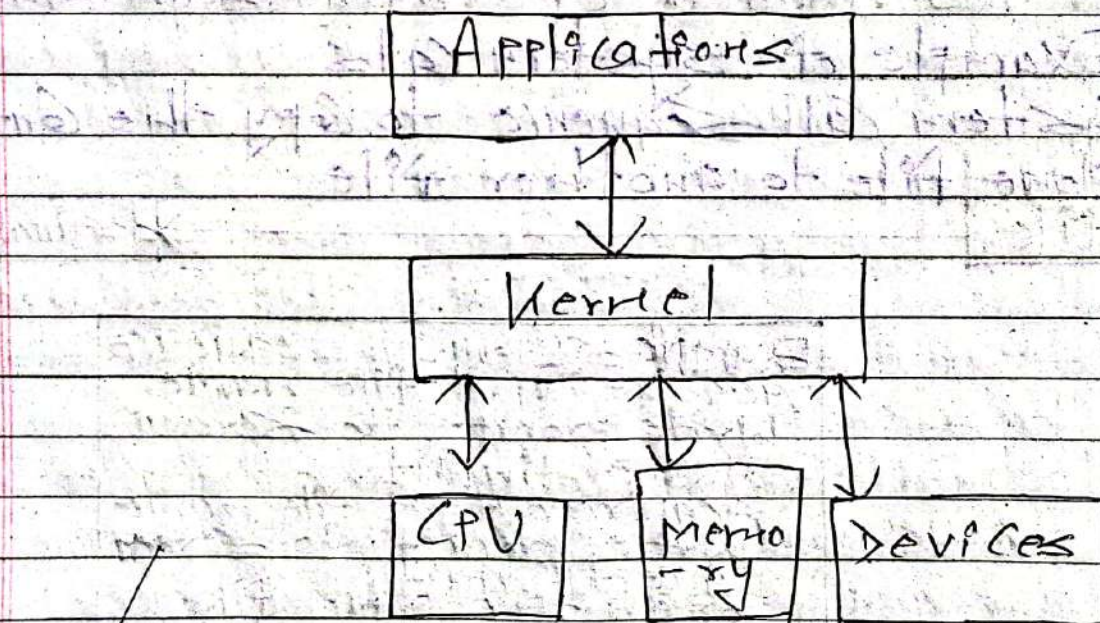
- Shell and the Kernel are the parts of the operating system. These both are used for performing any operation on the system.
- When a user gives his command for performing any operation, then the request will go to the shell part. The shell part is also called as the 'interpreter' which translates the human program into the machine language and then the request will be transferred to the Kernel. That means shell is just as the interpreter of the commands which converts the request of the user into the machine language.
- Kernel is also known as the heart of the operating system and every operation is performed by using the kernel. When the kernel receives the request from the shell then this will process the request and display the results on the screen.
- The functions of Kernel are:
 - x) It controls the state of process i.e. it checks whether the process is running or is waiting for the request of the user.

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*> It provides memory for the processes those are running on the system i.e. kernel runs the allocation and deallocation process.

*> It also maintains a time-table for all the process those are running.

→ kernel is a bridge between applications and the actual data processing done at the hardware level.



Evolution of operating system

1. Serial processing
2. Batch processing
3. Multiprogramming
4. Multitasking or time sharing system
5. Network operating system
6. Distributed operating system
7. Multiprocessor operating system
8. Real time operating system
9. Modern operating system.

Example of system calls

System call sequence to copy the contents of one file to another file

Source file

Destination file

Example system call sequence

- Acquire input file name
- Write prompt to screen
- Accept input
- Acquire output file name
- Write prompt to screen
- Accept input
- Open the input file
- If file doesn't exist, abort
- Create output file
- If file exists, abort
- Loop
- Read from the input file
- Write to output file
- Until read fails
- Close output file
- Write completion message to screen
- Terminate normally

Fig: Example of system call